

Aria Owners Manual (2024)

Introduction and model identification

This manual covers the 2024 THOR Aria 24 and Aria 28 travel trailers. Both models share the same frame design, appliance package, and electrical layout. They differ in overall length, tank capacities, and weight ratings. Keep this manual in the trailer. Warranty service must be performed by an authorized THOR dealer, and the dealer will ask for the full vehicle identification number (VIN) at every visit.

The VIN is stamped on the A-frame tongue on the driver side and printed on the federal certification label just inside the entry door. The first seven characters identify the model:

Model	VIN prefix	Overall length	Dry weight	GVWR	Cargo capacity
Aria 24	1THRA24	27 ft 4 in	5,900 lb	7,600 lb	1,700 lb
Aria 28	1THRA28	31 ft 8 in	7,100 lb	9,200 lb	2,100 lb

The eighth character is the model year code (R for 2024). Characters nine through seventeen are the production sequence. Production plant and build date appear on the certification label as MM/YYYY. Service bulletins refer to build dates, so note yours on the inside cover of this manual.

Never load the trailer beyond its GVWR. Cargo capacity is reduced by the weight of any dealer-installed options, water in the tanks (8.3 lb per gallon), and propane (4.2 lb per gallon).

Warranty

THOR warrants the Aria against defects in materials and workmanship for 36 months from the date of delivery to the first retail owner. The 36 month period begins on the delivery date recorded by the dealer on the warranty registration, not the purchase or build date.

What is covered

The structural warranty covers the welded steel frame, aluminum wall framing, laminated sidewalls, roof structure, and floor decking. The component warranty covers THOR-installed appliances, plumbing, the 12V and 120V electrical systems, slide-out mechanisms, the awning, and interior cabinetry. Appliances are serviced through THOR dealers; the owner does not need to contact the appliance maker directly.

What is excluded

The warranty does not cover tires, brake linings, light bulbs, fuses, sealant maintenance, or damage caused by overloading, improper hitching, water intrusion from neglected seals, freezing, or modifications. The AGM house battery (SKU BAT-AGM-100) carries a shorter 12 month warranty. Normal wear items such as slide-out wiper seals and awning fabric are covered for the first 12 months only.

Claim process

All claims go through an authorized dealer. The owner brings the trailer to the dealer, the dealer diagnoses the fault and records the VIN, mileage, and delivery date, and then submits a claim through the THOR dealer portal. Repairs with a total cost above 500 dollars require pre-authorization from THOR before work begins. The dealer must submit the completed claim within 30 days of the repair date. THOR pays approved claims within 45 days. Claims filed after the 36 month period are rejected unless the fault was reported to the dealer before the period ended.

Hitching and towing

The Aria uses a conventional ball hitch. Both models require a 2-5/16 inch hitch ball rated for at least the trailer GVWR. A 2 inch ball will appear to seat but will not latch correctly and can separate under load.

Target tongue weight is 10 to 12 percent of the loaded trailer weight. For a fully loaded Aria 24 this is roughly 760 to 910 lb; for the Aria 28 it is roughly 920 to 1,100 lb. Tongue weight below 10 percent causes sway. Weigh the loaded trailer at a certified scale after the first trip and adjust cargo placement.

A weight distribution hitch with integrated sway control is required for the Aria 28 and strongly recommended for the Aria 24. THOR does not specify a brand. Set the spring bars so the tow vehicle front axle returns to within 50 percent of its unhitched height change. Friction sway control bars must be released before backing.

Before every trip: check the coupler latch and safety pin, cross the safety chains under the coupler, plug in the seven-pin connector, test the running lights and brake lights, confirm the breakaway cable is attached to the tow vehicle and not to the hitch, and check tire pressure. The trailer tires are ST225/75R15 load range E on both models and run at 80 psi cold.

The electric brakes on all four wheels are controlled by a brake controller in the tow vehicle. Set the controller gain so a firm stop from 25 mph does not lock the trailer wheels. Adjust the brake shoes at 3,000 miles and every 12 months after that.

Slide-out operation

The Aria 24 has one slide-out (living room). The Aria 28 has two: the living room slide and the rear bedroom slide. Each slide is driven by a 12V rack-and-pinion motor, SKU SLD-MTR-12, mounted under the floor at the inboard end of the room.

The slide switch is next to the entry door. Hold the switch until the room stops and the motor clicks off. Do not release early; the seals only compress fully at the end of travel. The trailer must be level and the stabilizer jacks lowered before extending a slide. Keep people and pets clear of the room during travel.

Seal care

Each slide has an outer bulb seal that compresses against the room when retracted and a wiper seal that sweeps the roof of the room during travel. Clean both seals every 3 months with mild soap and water and treat with a dry silicone spray. Never use petroleum-based lubricants; they swell the rubber. Inspect the rear bedroom slide seal on the Aria 28 for cracks or loss of bulb shape at every 6 month service. A seal that stays flattened after the room is extended has lost its memory and should be replaced.

Manual override

If the motor fails, the slide can be moved by hand. Turn off the 12V disconnect. Locate the override hex on the motor gearbox under the floor. Remove the drive pin and turn the hex with a 3/4 inch socket on a ratchet. It takes about 40 turns to retract a slide fully. Reinstall the drive pin after the room is closed. Do not tow with a slide partially extended.

Water system

Model	Fresh water	Gray water	Black water
Aria 24	44 gal	38 gal	30 gal
Aria 28	52 gal	46 gal	38 gal

The water pump is a 12V diaphragm pump rated at 3.0 gallons per minute with a 55 psi cutoff. City water enters through a pressure regulator; the inlet must not exceed 60 psi.

Water heater

The water heater is a THOR AquaLine WH-10, a 10 gallon gas and electric unit mounted behind the exterior panel on the passenger side. The gas burner is 12,000 BTU; the electric element is 1,400 watts. Both can run at the same time for the fastest recovery. The gas side uses a direct-spark igniter, SKU WH-IGN-10, that fires three times before locking out. A lockout is indicated by a steady red light on the interior switch. Reset by turning the switch off for 10 seconds and back on. If the heater locks out three times in a row, the igniter electrode gap (which should be 1/8 inch) or the gas supply pressure (11 inches water column) is likely at fault.

Never run the electric element with the tank empty. The element burns out in under a minute without water.

Winterizing

1. Drain the fresh tank using the low-point drain valve under the trailer.
2. Open the water heater bypass valve behind the heater and drain the heater through its anode rod port. Inspect the anode; replace it when less than 25 percent remains.
3. Open all faucets and the low-point hot and cold drains until flow stops.
4. Connect the winterizing hose from the pump to a jug of RV antifreeze and run the pump until each faucet, the toilet, and the outside shower discharge pink.
5. Pour one cup of antifreeze into each drain trap.
6. Close the bypass valve in the spring before refilling, or the heater will not fill.

Two gallons of antifreeze is enough for the Aria 24; the Aria 28 needs three.

Electrical

12V system

House power comes from one Group 31 AGM battery, SKU BAT-AGM-100, rated at 100 amp hours. A second battery can be added in parallel on the tongue tray. The battery is charged by the converter when connected to shore power or by the tow vehicle through the seven-pin connector while driving. A battery disconnect switch is located in the pass-through storage. Turn it off during storage; the propane detector and stereo memory otherwise draw about 0.4 amps and will flatten the battery in about 10 days.

Battery state of charge can be estimated from resting voltage: 12.8 volts is full, 12.4 volts is about 75 percent, 12.0 volts is about 50 percent. Do not discharge the AGM battery below 12.0 volts.

Converter and inverter

The converter is a 55 amp three-stage unit in the fuse panel cabinet. It puts out 14.4 volts in bulk mode, 13.6 volts in absorption, and 13.2 volts in float. A converter stuck in bulk mode for more than 24 hours will overcharge the battery.

An inverter, SKU INV-2000, is available as a dealer-installed option. It is a 2,000 watt pure sine wave unit that powers the two living-area outlets and the television when not on shore power. It draws up to 180 amps from the battery at full output, so a single BAT-AGM-100 will run it for less than 30 minutes at full load. THOR recommends two batteries when the INV-2000 is fitted.

Fuse panel

The 12V fuse panel is under the refrigerator. Fuses are ATO blade type. Circuits: 15 amp furnace, 15 amp water pump, 10 amp water heater, 20 amp slide-out control, 15 amp interior lights, 10 amp propane detector and stereo, 30 amp awning, 15 amp tank heaters. The 120V panel has a 30 amp main breaker and 15 amp branch breakers for the converter, GFCI outlets, water heater element, and air conditioner (20 amp).

Furnace and appliances

Furnace

The furnace is a THOR ThermaCore TC-30 rated at 30,000 BTU. It is a ducted forced-air unit mounted under the dinette. It requires at least 10.5 volts to run; below that the blower slows and the sail switch will not close. The sail switch, SKU FRN-SAIL, is a paddle switch inside the blower housing that proves airflow before the gas valve opens. A dusty or bent sail switch is the most common cause of a furnace that blows air but never lights.

The furnace control board flashes a fault code on the red LED visible through the intake grille:

Code Meaning

Typical cause

F1	Sail switch did not close within 30 seconds	Low voltage, blocked return air, bent or dirty FRN-SAIL
F2	High limit switch open	Blocked ducts, closed registers, failed limit switch
F3	Ignition lockout after 3 attempts	Empty propane, air in line, fouled electrode
F4	Flame sensed with gas valve closed	Faulty gas valve or board
F5	Supply voltage below 10.5 volts	Discharged battery, corroded connector

Reset any code by cycling the thermostat off for 5 minutes. Keep the furnace exhaust on the exterior wall clear of snow and debris.

Refrigerator, range, and air conditioner

The refrigerator is a 10 cubic foot 12V compressor unit. It draws about 4 amps when running and should be started 8 hours before loading. The range has three burners and a 17 inch oven. The oven pilot is lit by hand; hold the knob in for 30 seconds after the pilot lights. The roof air conditioner is 13,500 BTU and requires shore power or a 3,000 watt generator; it will not run from the INV-2000.

Awning

The power awning uses a 12V motor, SKU AWN-1200, in the roller tube. The switch is inside the entry door. Extend the awning fully; a partially extended awning flaps more in wind. Lower one end by 6 inches with the arm adjuster so rain runs off.

Retract the awning whenever wind exceeds 20 mph or when leaving the trailer. Gusts above 30 mph can bend the arms even when the awning is retracted only partway. Wind damage is not covered by warranty. In the event of motor failure, remove the plastic cap on the driver-side arm and turn the manual crank hex with a 7/16 inch socket. Clean the fabric twice a year with a soft brush and diluted dish soap; allow it to dry before rolling.

Maintenance schedule

Item	Interval
Check tire pressure (80 psi) and lug nut torque (110 ft-lb)	Before every trip
Test propane detector and smoke alarm	Monthly
Clean and treat slide-out seals	Every 3 months
Inspect roof and sidewall sealant, reseal as needed	Every 6 months
Inspect slide-out seals for cracking	Every 6 months
Repack wheel bearings	12 months or 12,000 miles
Adjust electric brakes	3,000 miles, then every 12 months
Replace water heater anode rod	12 months or when 75 percent consumed
Clean furnace burner and sail switch	12 months
Load test house battery	12 months
Lubricate slide-out gear rack with dry lubricant	12 months
Replace propane regulator	60 months

Troubleshooting

Symptom	Likely cause	Action
Nothing works on 12V	Battery disconnect off or battery below 10 volts	Turn on disconnect; charge battery; check 30 amp main fuse
Slide-out will not move, motor hums	Trailer not level or room binding	Level trailer; check for obstruction; use manual override if needed
Slide-out moves in one direction only	Failed switch or motor relay	Dealer to test switch and replace SLD-MTR-12 if motor fails one direction

Water pump runs continuously	Air leak on suction side or empty tank	Fill tank; check pump strainer and fittings
Water heater red light on	Igniter lockout	Reset switch; check propane; check WH-IGN-10 gap
Furnace blows but no heat, code F1	Sail switch not closing	Charge battery; clear return air; clean or replace FRN-SAIL
Furnace cycles on and off, code F2	Restricted airflow	Open all registers; check duct connections
Awning will not retract	Motor stalled or fuse blown	Check 30 amp fuse; use manual crank; dealer to replace AWN-1200
Trailer sways at speed	Tongue weight below 10 percent	Move cargo forward; verify weight distribution setup
Refrigerator not cooling, fan runs	Low voltage or blocked condenser	Check battery voltage above 12.0; clear exterior vent
GFCI outlets dead	Tripped GFCI	Reset GFCI in bathroom; it feeds all exterior and kitchen outlets